CHAPTER 2

Holography

2.1 Introduction

The conventional photography records only the intensities of light coming from an object. It fails to record the phases of the waves that come from the objects. Hence, they show two-dimensional images of three-dimensional objects. In 1948, Dennis Gabor, a British scientist, developed a method of recording and producing three-dimensional images of objects through interference phenomena of coherent light known as holography. In Greek, 'holo' means 'whole' or 'complete', 'holography' means 'complete recording' i.e., the intensities and phase of the waves that come from the objects are recorded. In 1971, he received noble prize for his efforts.

2.2 Basic principle of holography

An object is illuminated with a beam of coherent light [object beam]. Then every point on the surface of the object acts as a source of secondary waves. These secondary waves spread in all directions. Some of these waves are allowed to fall on a recording plate [holographic plate]. Simultaneously, another beam of same coherent light [reference beam] is allowed to fall on this holographic plate. In the holographic plate, both the beams combine and interference pattern will be formed. This interference pattern is recorded on the holographic plate. The three-dimensional image of the object can be seen by exposing the recorded holographic plate [hologram] to coherent light. This is the principle of holography.

2.3 Recording of image on a holographic plate

Figure 2.1 shows the method of recording a image on a holographic plate. The monochromatic light from a laser has been passed through a 50% beam splitter so that the amplitude division of the incident beam into two beams takes place. One beam falls on mirror M_1 and the light reflected from M_1 falls on the object. This beam is known as an object beam.

The object scatters this beam in all directions, so that a part of the scattered beam falls on the holographic plate. The other beam is reflected by mirror M_2 and falls on the holographic plate. This beam is known as reference beam. Superposition of the scattered rays from the object and the reference beam takes place on the plane of the holographic plate, so that interference pattern is formed on the plate and it is recorded. The recorded interference pattern contains all the information of the scattered rays i.e., the phases and intensities of the scattered rays. For proper recording, the holographic plate has to be exposed to the interference pattern for a few seconds. After exposing, the holographic plate is to be developed and fixed as like in the case of ordinary photograph. The recorded holographic plate is known as hologram or Gabor zone plate. The hologram does not contain a distinct image of the object. It contains information in the form of interference pattern.

2.4 Reconstruction of image from a hologram

As shown in Fig. 2.2, the hologram is exposed to the laser beam [that used during construction or identical to the reference beam used for construction] from one side and it can be viewed from the other side. This beam is known as reconstruction beam. The reconstruction beam illuminates the hologram at the same angle as the reference beam. The hologram acts as a diffraction grating, so constructive interference takes place in some directions and destructive interference takes place in other directions. A real image is formed in front of the hologram and a virtual image is formed behind the hologram. It is identical to the object and hence it appears as if the object is present. The three-dimensional effect in the image can be seen by moving the head of the observer. During recording, the secondary waves from every point of the object reach complete plate. So, each bit of the plate contains complete information of the object. Hence, image can be constructed using a small piece of hologram.

2.5 Comparison between holography and photography

Holography	Photography
(i) In holography, the three dimensional image of an object can be recorded.	(i) In photography, the two dimensional image of an object can be recorded.
(ii). The intensities and the phases of waves scattered from objects are recorded.	(ii) The intensities of the scattered waves from objects only recorded but the phases of waves are not recorded.
(iii) Large number of images have been recorded in a single holographic plate.	(iii) Single image of an object is recorded in a photographic plate
(iv) A hologram contains recorded interference pattern due to the scattered waves from the object and reference monochromatic beam but not the image of the object	(iv) A photographic plate contains the recorded image of an object

Figure 2.1 Recording of hologram

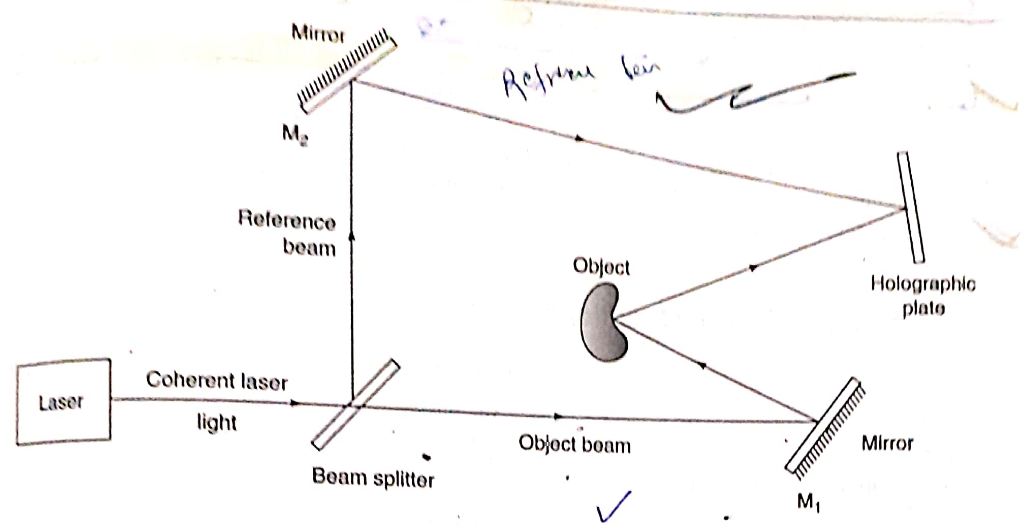
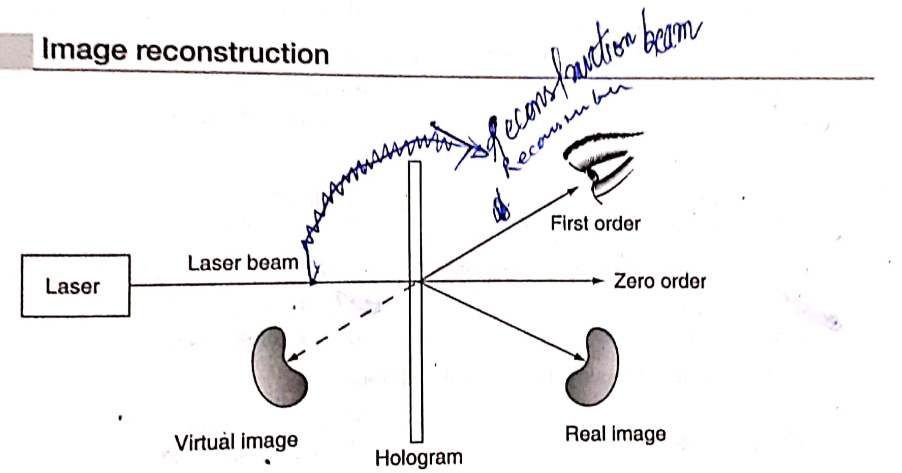


Figure 2.2 Image reconstruction



2.6 Applications of holography

1. The three-dimensional images produced by holograms have been used in various fields, such as technical, educational also in advertising, artistic display etc.
2. Holographic diffraction gratings: The interference of two plane wavefronts of laser beams on the surface of holographic plate produces holographic diffraction grating. The lines in this grating are more uniform than in case of conventional grating.
3. Hologram is a reliable object for data storage, because even a small broken piece of hologram contains complete data or information about the object with reduced clarity.

4. The information-holding capacity of a hologram is very high because many objects can be recorded in a single hologram, by slightly changing the angle between reference beam and holographic plate. For each different angle, different images can be stored.
5. In hospitals holography can be used to view the working of inner organs three dimensionally i.e., the beating of the heart, the foetus of the pregnant lady and flowing blood based on motion holography.
6. Holographic interferometry is used in non-destructive testing of materials to find flaws in structural parts and minute distortions due to stress or vibrations, etc. in the objects.
7. Holography is used in information coding.